

SEEHAUSEN ALTMARK, Germany

WMO: 102610

Lat: 52.8911N

Lon: 11.7297E

Elev: 23

StdP: 101.05

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.5	-10.6	-15.9	0.9	-12.4	-13.0	1.2	-10.2	12.3	5.8	10.8	5.3	1.2	110	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	29.6	18.8	27.6	18.0	25.7	17.3	20.2	26.3	19.1	25.0	18.2	23.8	3.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.1	22.0	17.1	12.2	21.1	16.0	11.4	20.4	57.9	26.0	54.4	25.2	51.4	23.5	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.3	8.9	7.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-14.9	33.0	6.3	2.0	-19.4	34.4	-23.1	35.6	-26.6	36.7	-31.2	38.1
			-15.1	22.5	6.2	1.1	-19.6	23.3	-23.2	24.0	-26.7	24.6	-31.2	25.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	0.5	0.2	4.2	7.5	12.7	15.6	17.8	17.5	13.9	9.5	4.4	2.1	(6)
(7)		DBStd	7.38	5.89	5.19	3.76	3.48	3.55	3.37	3.04	2.96	2.68	3.70	3.63	4.03	(7)
(8)		HDD10.0	1327	296	276	180	90	14	1	0	0	2	53	169	247	(8)
(9)		HDD18.3	3529	553	509	437	324	178	96	47	52	137	274	418	504	(9)
(10)		CDD10.0	917	1	1	2	16	97	168	242	233	118	38	2	1	(10)
(11)		CDD18.3	78	0	0	0	0	2	14	32	27	3	1	0	0	(11)
(12)		CDH23.3	1016	0	0	0	2	67	207	400	303	36	3	0	0	(12)
(13)		CDH26.7	269	0	0	0	0	5	55	117	89	3	0	0	0	(13)
(14)	Wind	WSAvg	3.4	4.1	3.6	3.7	3.3	3.1	3.1	3.0	2.8	3.1	3.2	3.6	3.8	(14)
(15)	Precipitation	PrecAvg	583	49	37	38	31	49	57	71	62	51	45	46	48	(15)
(16)		PrecMax	786	112	75	97	75	129	98	142	145	138	123	99	82	(16)
(17)		PrecMin	407	2	11	7	2	12	21	16	16	11	12	2	19	(17)
(18)		PrecStd	95	24	16	22	17	29	23	34	31	28	26	23	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	14.5	17.3	22.4	27.2	31.6	31.7	31.9	26.9	22.2	13.5	12.4	(19)
(20)			MCWB	9.4	11.1	10.9	13.7	16.4	18.7	20.1	20.9	17.5	16.6	10.8	10.2	(20)
(21)		2%	DB	9.7	11.1	13.9	19.1	25.3	28.1	29.7	29.1	24.0	19.1	12.1	10.9	(21)
(22)			MCWB	8.0	8.6	9.8	12.1	15.4	17.6	19.0	18.9	17.1	14.7	10.2	9.4	(22)
(23)		5%	DB	8.5	8.4	11.4	16.7	23.2	25.6	27.8	26.8	21.3	16.7	11.0	9.5	(23)
(24)			MCWB	7.0	6.6	8.1	10.8	14.7	16.5	18.5	18.3	15.6	13.0	9.5	8.3	(24)
(25)		10%	DB	7.2	6.5	9.8	14.2	20.9	23.3	25.5	24.3	19.3	15.0	9.6	8.2	(25)
(26)			MCWB	5.8	4.9	7.4	9.6	13.8	16.0	17.8	17.3	14.8	12.2	8.3	7.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	11.3	11.9	14.5	18.0	20.9	21.8	21.6	19.1	17.5	11.6	10.7	(27)
(28)			MCDWB	11.3	13.9	16.3	21.1	24.1	24.6	28.8	29.5	24.7	21.3	12.9	11.8	(28)
(29)		2%	WB	8.5	8.6	10.2	12.7	16.6	18.9	20.5	19.9	17.8	15.3	10.6	9.6	(29)
(30)			MCDWB	9.5	10.7	13.2	17.6	23.3	25.3	27.1	25.6	22.4	18.3	11.8	10.7	(30)
(31)		5%	WB	7.2	6.8	8.9	11.4	15.4	17.7	19.2	19.0	16.7	13.7	9.6	8.3	(31)
(32)			MCDWB	8.2	8.3	10.8	15.5	21.7	23.9	25.7	24.2	20.3	16.0	10.8	9.3	(32)
(33)		10%	WB	5.7	5.0	7.5	10.2	14.3	16.5	18.1	18.1	15.4	12.5	8.4	7.1	(33)
(34)			MCDWB	7.0	6.3	9.6	14.0	19.9	22.1	24.0	23.2	18.6	14.5	9.5	8.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	6.3	7.3	9.9	11.3	10.5	11.1	10.8	9.3	7.9	5.3	4.3	(35)
(36)			MCDBR	5.4	7.8	10.7	14.5	15.9	15.8	16.1	15.5	13.6	10.8	6.6	5.5	(36)
(37)			MCWBR	4.6	6.0	7.2	8.1	7.9	7.0	7.2	7.2	7.6	6.9	5.4	4.9	(37)
(38)		5% WB	MCDWR	5.2	7.6	9.9	12.6	13.8	13.2	14.0	12.4	11.7	9.6	6.3	5.2	(38)
(39)			MCWBR	4.7	6.2	7.2	7.6	7.5	6.5	7.2	6.4	7.1	6.7	5.4	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.318	0.340	0.376	0.413	0.411	0.408	0.428	0.420	0.386	0.362	0.338	0.311	(40)	
(41)		taud	2.339	2.326	2.251	2.201	2.254	2.282	2.241	2.274	2.353	2.396	2.382	2.337	(41)	
(42)		Ebn at Noon	659	758	800	816	839	844	817	801	786	725	630	594	(42)	
(43)		Edh at Noon	60	82	108	128	128	125	129	118	97	76	58	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.62	1.31	2.42	4.00	4.96	5.29	4.96	4.19	3.07	1.71	0.76	0.47	(44)	
(45)		RadStd	0.07	0.19	0.24	0.51	0.59	0.41	0.69	0.34	0.35	0.20	0.08	0.04	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	+0.46	N/A	-150	N/A	N/A

Nomenclature: See separate page